ing the standards of care offered to persons suffering from diabetes, and integrating the disease into the general health care scheme—the Inter-Health approach.

The importance of coordinated action cannot be overemphasized. For instance, there is an increasing need for human and financial resources which the private sector is eager to offer. Major sponsorship by large industrial enterprises and bodies such as the Lions and Rotary Clubs has stimulated crucial advances in the field of diabetes control.

One of the many initiatives taken to pursue WHO's objective began in Italy in 1989, when the European regional offices of both the IDF and WHO met to assess the diabetes situation, and enshrined their common goals in a single document. This is known as the St Vincent Declaration, whose subtitle expresses the wide scope of action to be undertaken: "Diabetes mellitus in Europe: a problem at all ages, in all countries: a model for prevention and self-care".

Future action

Researchers, clinicians, volunteers and delegates from 98 diabetes associations, as well as persons suffering from diabetes, will convene in Washington D.C. in June 1991 to participate in the 14th Congress of the Federation. World experts, representatives of nongovernmental organizations and the private sector, as well as delegates of government agencies, universities and medical schools, all of them committed to meet the challenges of the care of diabetics and research in diabetes, will use this occasion to further the global efforts to conquer the disease in the near future.

The International Diabetes Federation believes it has made a positive response to the growing need for action. It values the trust it has received, and is grateful to WHO, its Member States and its Director-General, Dr Hiroshi Nakajima, for recognizing the importance of diabetes. The Federation and its members worldwide will urge governments, universities and the private sector to work with intelligence, devotion and humanity in order to improve the lives of persons with diabetes throughout the world. ■

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What is diabetes?

Diabetes mellitus is gaining ground throughout the world, in both developing and developed countries. It is a chronic disease caused by inherited or acquired deficiency in the production of insulin by the pancreas, which results in an increased concentration of sugar in the blood. Insulin-dependent diabetes appears early in life and can usually be regarded as hereditary, even though the mechanism by which it passes from one generation to the next has not yet been fully explained. Children who have two diabetic parents run a 20% risk of becoming diabetic themselves.

Another type of diabetes, which is noninsulin-dependent, appears later in life, is often concomitant with obesity and is mainly caused by inappropriate diet.

The classic symptoms of diabetes are excessive secretion of urine (polyuria), great thirst (polydipsia), weight loss in spite of excessive appetite (polyphagia), and a feeling of lassitude. Sometimes no early symptoms appear, and the disease is only diagnosed when routine examination shows sugar in the urine or a raised blood sugar level.

Well-treated diabetic individuals do not show any outward signs of the disease and should be regarded as in good health. However, excessive sugar in the blood vessels leads to vascular problems, and sooner or later complications are liable to occur in the circulatory system.

Eye impairment is one of the vascular complications that diabetics fear most. The disease is a major cause of blindness in the world, even though there are few diabetic persons who completely lose their sight. It also produces reduced skin sensitivity that may lead to severe ulcerations in the limbs and eventually bring about another very severe complication: gangrene leading to amputation.

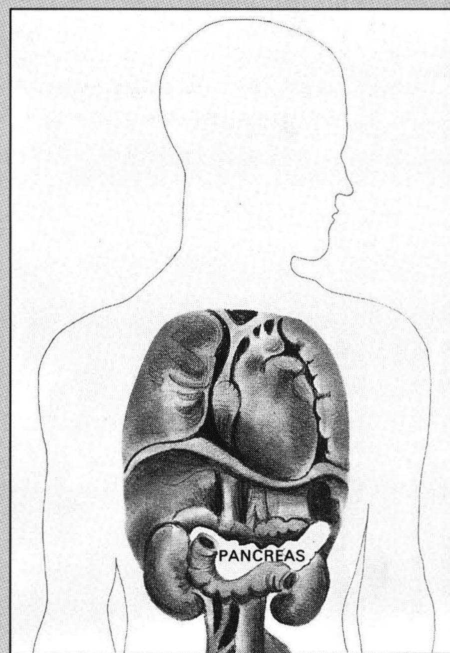
Insulin

Insulin is an essential treatment for subjects with insulin-dependent diabetes and is also required by a proportion of noninsulin-dependent diabetic individuals. It compensates for the lack of endogenous insulin resulting from defective functioning of the pancreas. Insulin is traditionally administered through injections, mostly twice a day, usually given by the patients themselves; at present 25–30% of diabetics require this kind of treatment. New technology, e.g., the insulin pump, is attempting to bring increased comfort and improved metabolic control, with the computer-controlled miniaturized system constantly monitoring the patient's needs and injecting insulin as required. This is particularly the case with young patients, as those who contract the disease in adulthood often retain a residual capacity for producing insulin.

Genetic engineering is expected to make it increasingly easy to produce large quantities of high-quality insulin through bacterial culture.

Education is indispensable

Diet, physical exercise, and drugs are three cornerstones of diabetes treatment; the fourth is education.



Drawing by Peter Davies

A deficiency of the insulin-producing mechanism in the pancreas lies at the origin of diabetes.

A strict diet is essential, since even with insulin treatment the blood sugar level can only be kept within reasonable limits by controlling food intake. In many cases obesity leads to diabetes, and in fact most people who contract diabetes in adulthood are overweight. For many of them, a simple reduction in weight can make other treatment superfluous. Avoiding and treating obesity will reduce the risk of diabetes in older people with a family history of diabetes, and in women who have given birth to babies weighing more than 4.5 kg at birth.

Education is just as important for diabetics as diet, drugs and exercise, since treatment cannot be effective unless the patient understands the disease and takes positive steps to combat it. Thanks to the acceptance of these basic principles, the life expectancy of diabetic individuals in some countries is now practically the same as that of the general population.

C.V.